SRM IST
School of Computing
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21CSC315J - FOG COMPUTING

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Unit 5 - Application and Issues

- ❑Fog Computing Realization for Big Data Analytics,
- ❑Data Analytics in the Fog,
- ❑Prototypes and Evaluation.
- ❑Case Study 1: Fog Computing in E-Health Monitoring.
- ❑Case Study 2: Intelligent Traffic Lights Management (ITLM) System

CO5: Experiment the knowledge of Fog in the design of various application

1. Fog Computing Realization for Big Data Analytics

Introduction

- The impact of the Internet of Things (IoT) on data generation and the subsequent need for effective data processing and analytics.
- IoT devices produce vast amounts of data in real-time, necessitating advanced infrastructure and methodologies for analysis.

Key Points

- **Data Volume and Velocity:** The increasing number of IoT devices and applications leads to an unprecedented volume and velocity of data.
- **Importance of Data Processing:** There is a critical need for real-time data processing to generate actionable insights from the incoming data streams.

Challenges Faced by Current IoT Systems

1. High Overhead:

- Data movement on virtualized platforms incurs significant overhead which affects performance in terms of time, throughput, energy consumption, and cost.

2. Latency Issues:

- The physical distance from cloud data centers creates latency problems, making it challenging to process data quickly.

3. Increased Load on Data Centers:

- As data generation increases, it puts additional strain on data center resources, leading to potential security vulnerabilities and capacity constraints.

4. Cloud Capability Limitations:

- Traditional cloud platforms may struggle to effectively manage and analyze big data, particularly when demands for analytics are high.

5. Platform Fragmentation:

- The lack of standardization among IoT development

2. Big Data Analytics

Importance of Big Data Analytics

- **Definition:** Big data analytics refers to techniques and tools used to analyze large, diverse data sets to uncover patterns, correlations, and insights.
- **Applications:** Useful across various sectors, including finance, healthcare, and retail, to improve decision-making and optimize operations.
- **Competitiveness:** Organizations using analytics effectively can gain a competitive edge by anticipating market trends, understanding customer behavior, and improving operational efficiency.

2.1A Typical Big Data Analytics Infrastructure

- This subsection presents the foundational framework and components of a standard big data analytics infrastructure.

2.2.1 Big Data Platform

- **Framework:** A big data platform is necessary to integrate and manage data while providing analytics capabilities.
- **Key Components:**
  - **Hadoop Ecosystem:** Described as the cornerstone of big data platforms, allowing robust and scalable data analysis.
  - **Advantages:** Offers improved cost efficiency and performance over traditional database systems.

2.2.2 Data Management

- **Significance:** Proper data management and governance are vital to ensure high-quality data for analysis.
- **Processes Involved:**
  - Establish standard procedures to manage incoming and outgoing data.
  - Focus time on data cleaning, removing inaccuracies, and formatting data for analysis.
- **Master Data Management:** A strategy should be implemented to align data across the organization, ensuring unified quality and consistency.

2.2.3 Data Sources

- **Diversity of Data:** Data can be collected from numerous sources, including sensors, social media, transaction records, and more.
- **Integration:** Combining structured and unstructured data from various origins is crucial for comprehensive analysis.

2.2.4 Data Processing Engines

- **Role of Processing Engines:** Essential for executing complex queries and processing data for analysis.
- **Examples:**
- Use of tools like Apache Spark or Apache Flink for efficient data processing and real-time analytics.

2.2.5 Data Analytics Techniques

- **Methods:** Various techniques such as statistical analysis, machine learning, and data visualization are employed to derive insights from data.

2.3 Technologies

- **Integration Challenges:** Various technologies are indispensable to realizing effective big data analytics due to the size and variety of data. The integration of disparate tools and platforms is a significant challenge.
- **Data Silos:** Different storage platforms can create isolated silos of data, complicating the analytics process.
- **Technology Stack:** A cohesive architecture needs to be developed, often integrating existing tools like Hadoop with complementary technologies to meet specific organizational needs.
- **Collaboration:** Platform engineers and analytics teams must work closely to determine the right mix of technologies and successfully integrate them into

2.4 Big Data Analytics in the Cloud

- **Shift to Cloud:** There has been a transition from mostly on-premises big data systems to cloud deployments, particularly leveraging Hadoop.
- **Cloud Benefits:**
- Flexibility and scalability of cloud solutions enable organizations to process large datasets without investing heavily in physical infrastructure.
- Public clouds such as AWS (Amazon Web Services) and Microsoft Azure provide environments suitable for deploying Hadoop clusters.
- **Hybrid Solutions:** The future of big data analytics is expected to be a hybrid approach, combining on-premises

2.5 In-Memory Analytics

- **Need for Fast Processing:** Traditional Hadoop systems often struggle with real-time data analysis due to batch processing approaches that introduce latency.
- **In-Memory Technology:** This technology allows data to be processed faster by storing it in memory rather than on disk, significantly reducing access times and improving performance.
- **Job Scheduling:** In-memory analytics can begin processing jobs in milliseconds, in contrast to the slower startup times associated with traditional disk-based methods.
- **Examples:** Tools like SAS and Apache Ignite enhance Hadoop distributions by incorporating in-memory analytics features, thereby promoting quicker insights.

2.6 Big Data Analytics Flow

- **Overview:** The flow of big data analytics involves several key steps that facilitate transforming raw data into actionable insights.
- **Typical Flow Steps:**

1. **Data Collection/Integration:** Gather data from multiple sources and consolidate it for further analysis.
2. **Data Cleaning:** Essential to improve data quality by detecting and correcting errors, removing inconsistencies, and transforming data into a usable format. This step may be time-consuming but is crucial for ensuring reliable outcomes.
3. **Transformation:** Convert raw or unstructured data into structured or semi-structured formats to facilitate analysis.
4. **Feature Extraction:** Identify and extract relevant features from the cleaned data to prepare it for modeling.
5. **Data Mining & Analytics:** Apply analytical techniques to identify patterns, correlations, and trends within the data.

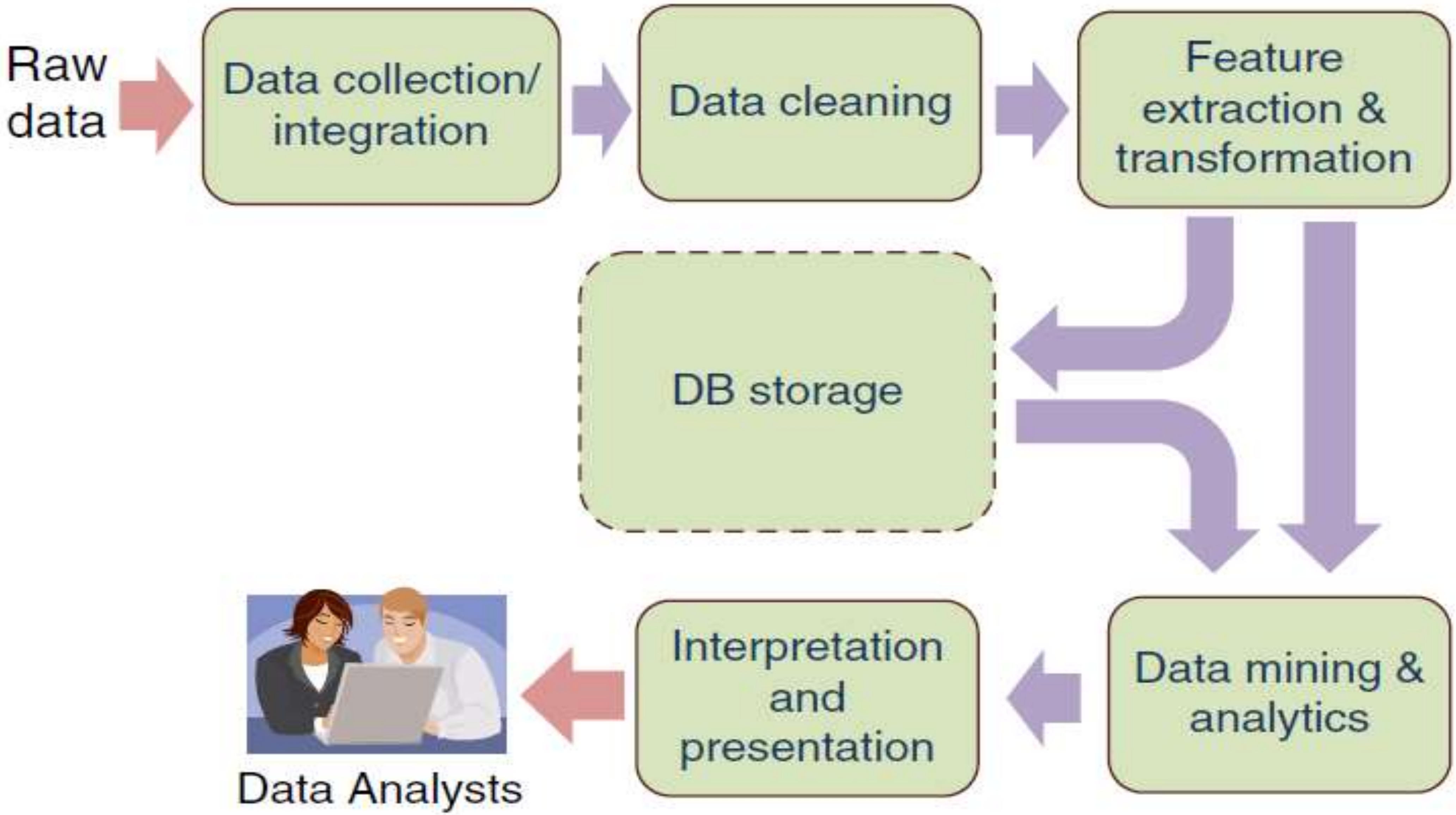


Figure 11.1 Typical data analytics flow.

3. Data Analytics in the Fog

- **Overview:** This section discusses the shift from traditional cloud-based data analysis to fog computing for better efficiency and responsiveness. Fog computing emphasizes processing data closer to where it is generated, particularly from IoT devices.
- **Challenges Addressed:** It highlights the limitations of cloud computing in handling real-time analytics due to latency issues and the costs associated with transmitting large amounts of data to central servers.
- **Advantages of Fog Computing:**
  - **Reduced Latency:** Processing occurs closer to the data source, allowing for quicker decision-making essential for real-time applications.
  - **Reduced Bandwidth Usage:** By processing data locally, less data is sent to the cloud, which helps conserve bandwidth

3.1 Fog Analytics

- **Local Processing:** Fog analytics refers to data analytics completed at the edge of the network, enabling analysis before data is sent to the cloud.
- **Real-time Processing:** This approach is crucial for applications requiring immediate insights, like those found in industrial automation, smart cities, and healthcare.
- **Interactivity:** Fog computing allows IoT devices to interact more effectively, enabling collaboration between devices to share insights and results in real time.
- **Decentralized Infrastructure:** A decentralized model allows for better resource utilization, as different fog nodes can handle various parts of the analytical workload, improving overall system responsiveness.

3.2 Fog-Engines

Introduction to Fog-Engines: Fog-engines represent a crucial component of fog computing architecture, designed to facilitate efficient data processing and analytics at the edge of the network.

- **Functionality**:
  - **On-premise Analytics**: Fog-engines provide capabilities for real-time data analysis and processing close to where the data is generated, which reduces reliance on cloud infrastructure.
  - **Inter-device Communication**: They enable interconnected IoT devices to communicate with one another, allowing for collaborative data processing and enhanced analytics.
- **Architecture**:
  - **Distributed Processing**: Fog-engines form a distributed network that allows them to share data and analytics results among themselves, effectively creating a local peer-to-peer network.
  - **Gateway Role**: Fog-engines act as gateways that facilitate the transfer of processed data to the cloud when necessary. This dual functionality bridges the on-premise and cloud-based analytics seamlessly.
- **Deployment Scenarios**:
  - **Standalone Operation**: A fog-engine can operate independently to preprocess and analyze data before it is sent to the cloud for further analysis.
  - **Cooperative Processing**: Multiple fog-engines can collaborate in a cluster to manage data from numerous IoT devices, optimizing storage

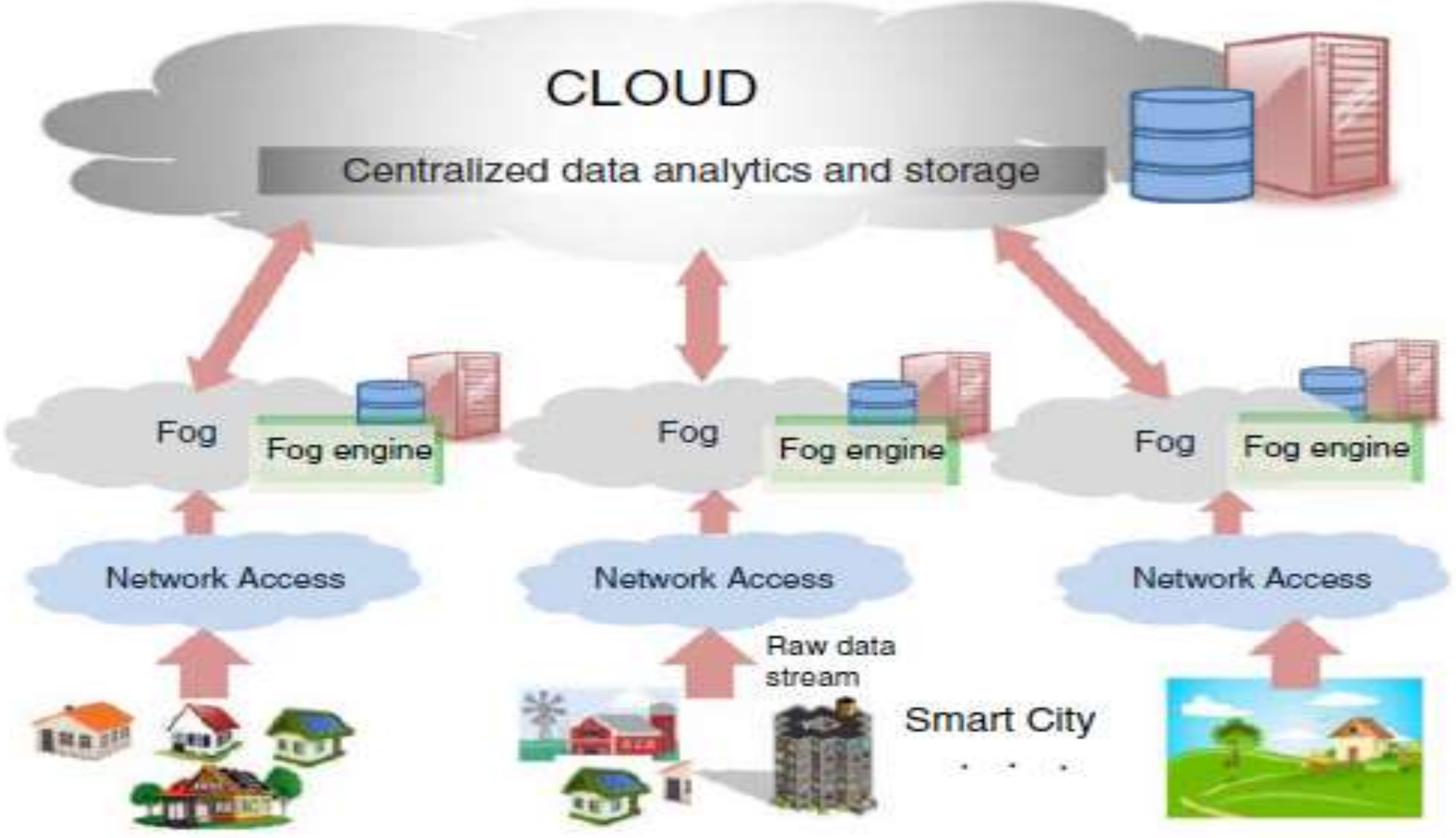


Figure 11.2 Deployment of FE in a typical cloud-based computing system.

3.3 Data Analytics Using Fog-Engines

Overview:

- This section discusses the framework of conducting data analytics at the edge of the network using fog-engines.
- It emphasizes how these engines can provide real-time, on-premise processing that enhances data handling efficiency.
- **Data Processing:**
  - **On-Premise Analytics:** Fog-engines analyze data locally at the source, allowing for immediate insights and reducing the volume of data that must be transmitted to the cloud.
  - **Preprocessing Steps:** The process includes data cleaning, feature extraction, and transformation. These steps are crucial to prepare the data adequately for analysis

- **Streamlined Data Flow:**
  - **Local Data Analysis:** Fog-engines perform analyses on incoming data streams in real-time to provide timely feedback and insights. For example, in a smart grid, the fog-engine can help users decide on efficient energy consumption.
  - **Cloud Engagement:** Data processed by fog-engines can be sent to the cloud for long-term storage and broader analytical tasks, allowing a hierarchical structure in data analysis where immediate insights are gained locally and larger trends are analyzed in the cloud.

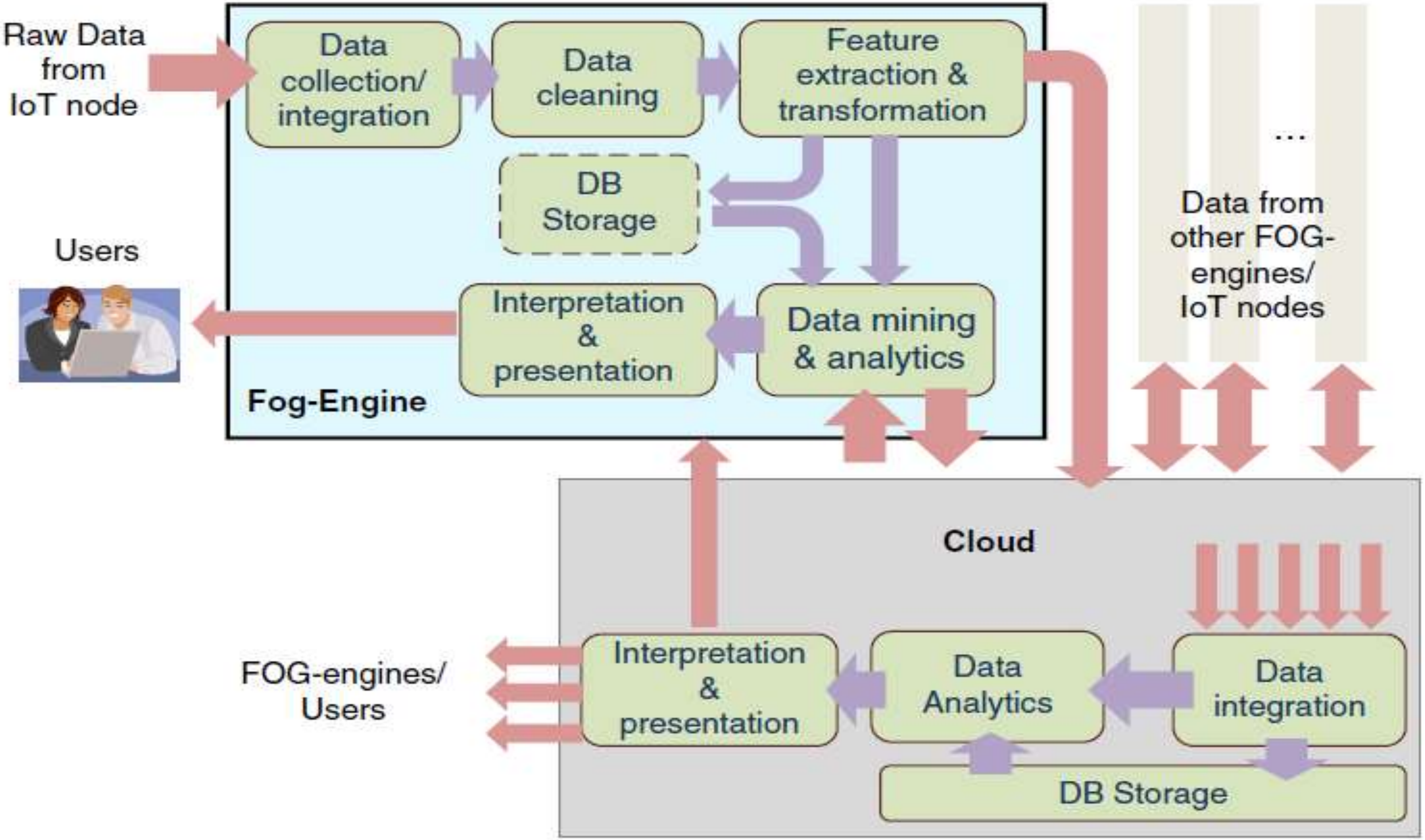


Figure 11.3 Data analytics using a fog-engine before offloading to the cloud.

System Architecture:

- **Integration with Other IoT Nodes:** Fog-engines act as hubs that facilitate data communications between connected devices within the IoT ecosystem.
- **Inter-fog Engine Communication:** Data from other fog-engines can be integrated, allowing for a collective analytical process that enhances overall system performance and insights.
- **Benefits of Using Fog-Engines:**
  - **Reduced Data Volume for Transmission:** By filtering and aggregating data at the edge, fog-engines significantly decrease the amount of data sent to the cloud, resulting in cost savings and improved efficiency.
  - **Immediate Analytics:** Real-time analytics allows for prompt actions to

- **Use Case Example:**
  - In applications such as smart cities, fog-engines can provide localized analytics for traffic management, enabling real-time adjustments to traffic signals or alerts about congestion, directly improving urban mobility.

- **Analytics Workflow:**
  - **Data Capture:** Raw data from IoT devices is captured continuously.
  - **Local Processing:** Upon collection, data is analyzed at the fog level where insights are generated immediately. Key tasks involve filtering out unnecessary data and identifying trends or anomalies.
  - **Final Transmission to Cloud:** Selected data insights, upon completion of local analysis, are then transmitted to the

Table: compares fog-engines with cloud computing

Table 11.1 Data analytics using a fog-engine and the cloud.

Characteristic	Fog-engine	Cloud
Processing hierarchy	Local data analytics	Global data analytics
Processing fashion	In-stream processing	Batch processing
Computing power	GFLOPS	TFLOPS
Network Latency	Milliseconds	Seconds
Data storage	Gigabytes	Infinite
Data lifetime	Hours/Days	Infinite
Fault-tolerance	High	High
Processing resources and granularity	Heterogeneous (e.g., CPU, FPGA, GPU) and Fine-grained	Homogeneous (Data center) and coarse-grained
Versatility	Only exists on demand	Intangible servers
Provisioning	Limited by the number of fog-engines in the vicinity	Infinite, with latency
Mobility of nodes	Maybe mobile (e.g. in the car)	None
Cost Model	Pay once	Pay-as-you-go
Power model	Battery-powered/ Electricity	Electricity

4. Prototypes and Evaluation

4.1 Architecture

- **Integration with IoT**: The fog-engine is designed for compatibility with low-end IoT devices, ensuring that it is:

1. Agile and transparent.
2. Non-disruptive to the existing IoT systems when integrated.

- **Composition of the Fog-Engine**: The fog-engine is divided into three primary units:

1. Analytics and Storage Unit:

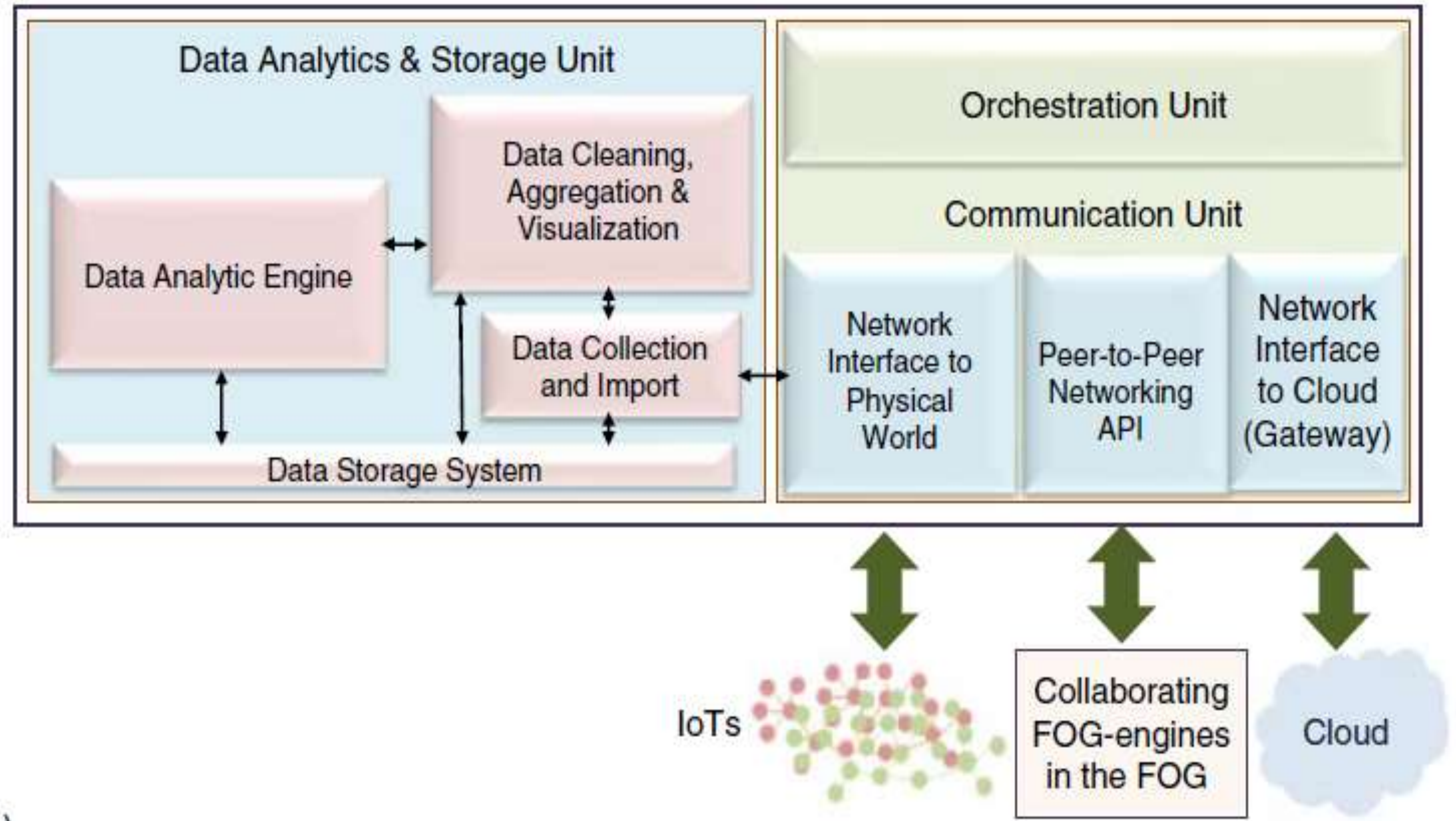
- Responsible for data preprocessing (cleaning, filtering).
- Handles data analytics and storage.

2. Networking and Communication Unit:

- Consists of network interfaces for peer-to-peer networking.
- Facilitates communication between fog-engine, IoT, and the cloud.

3. **Orchestrating Unit**: Ensures synchronization between fog-engines and the cloud.

- **Data Acquisition and Processing**:
  - The fog-engine interfaces with various data sources through:
  - USB, Wi-Fi for mid-range, Bluetooth for short-range communication, UART, SPI, and GPIO pins.
  - Collects data from sensors, IoT devices, the web, or local storage.
  - Implements preprocessing steps like cleaning, filtering, integration, and ETL (Extract, Load, Transform) procedures.
- **Data Rules and Logic**:
  - A rules library is used for data manipulation to ensure only acceptable data values are processed (e.g., positive values from smart meters).
- **Peer-to-Peer Networking**:
  - Preprocessed data can be shared with other fog-engines through peer-to-peer networking.
  - In a cluster of fog-engines, one with greater processing capability may act as a cluster head.
- **Cloud Communication**: A cloud interface module serves as a gateway for communication between the fog-engine and the cloud.


Data Analytics & Storage Unit
Data Cleaning, Aggregation & Visualization
Data Analytic Engine
Data Collection and Import
Data Storage System
Orchestration Unit
Communication Unit
Network Interface to Physical World
Peer-to-Peer Networking API
Network Interface to Cloud (Gateway)
IoTs
Collaborating FOG-engines in the FOG
Cloud
(a)

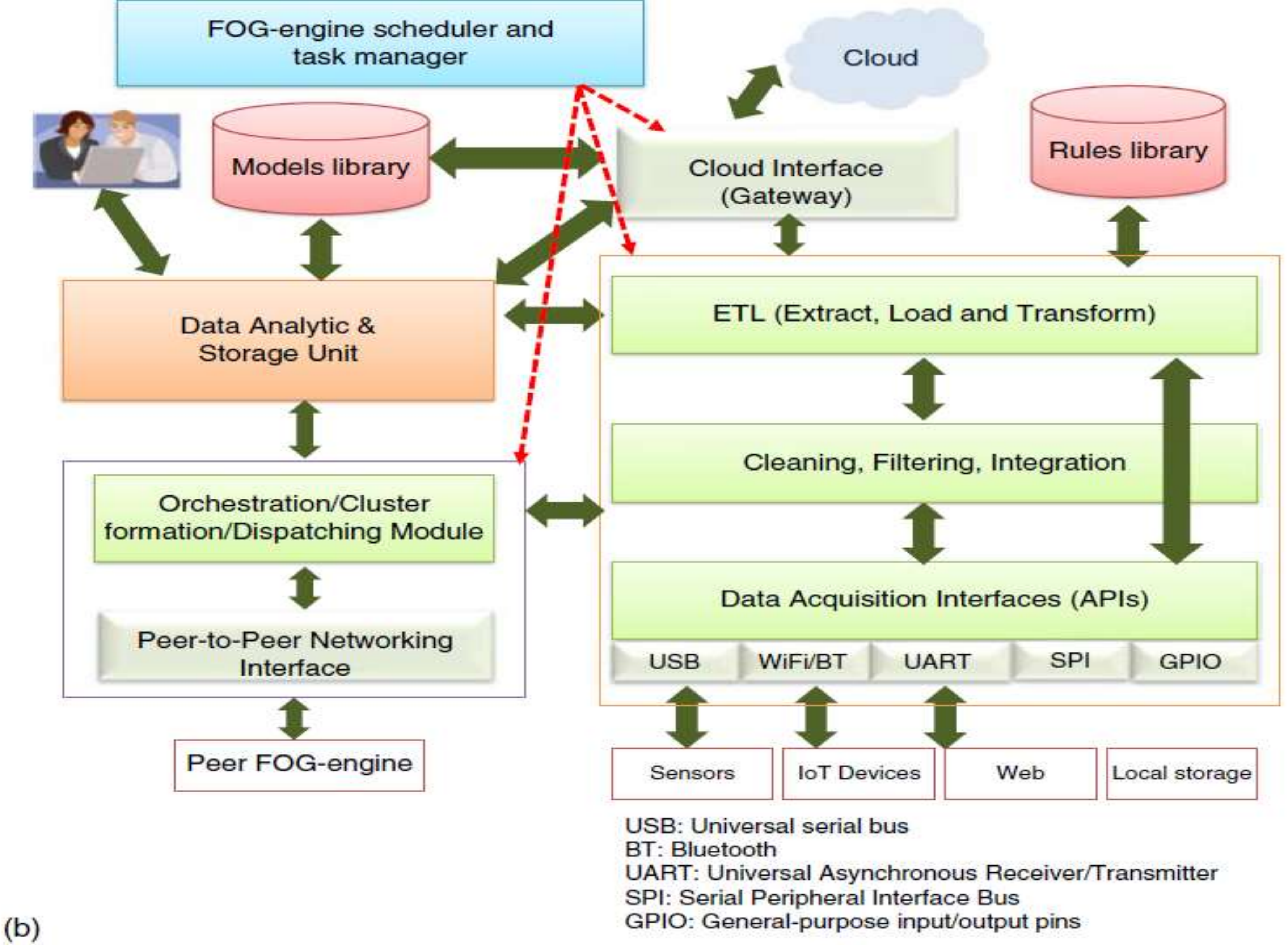


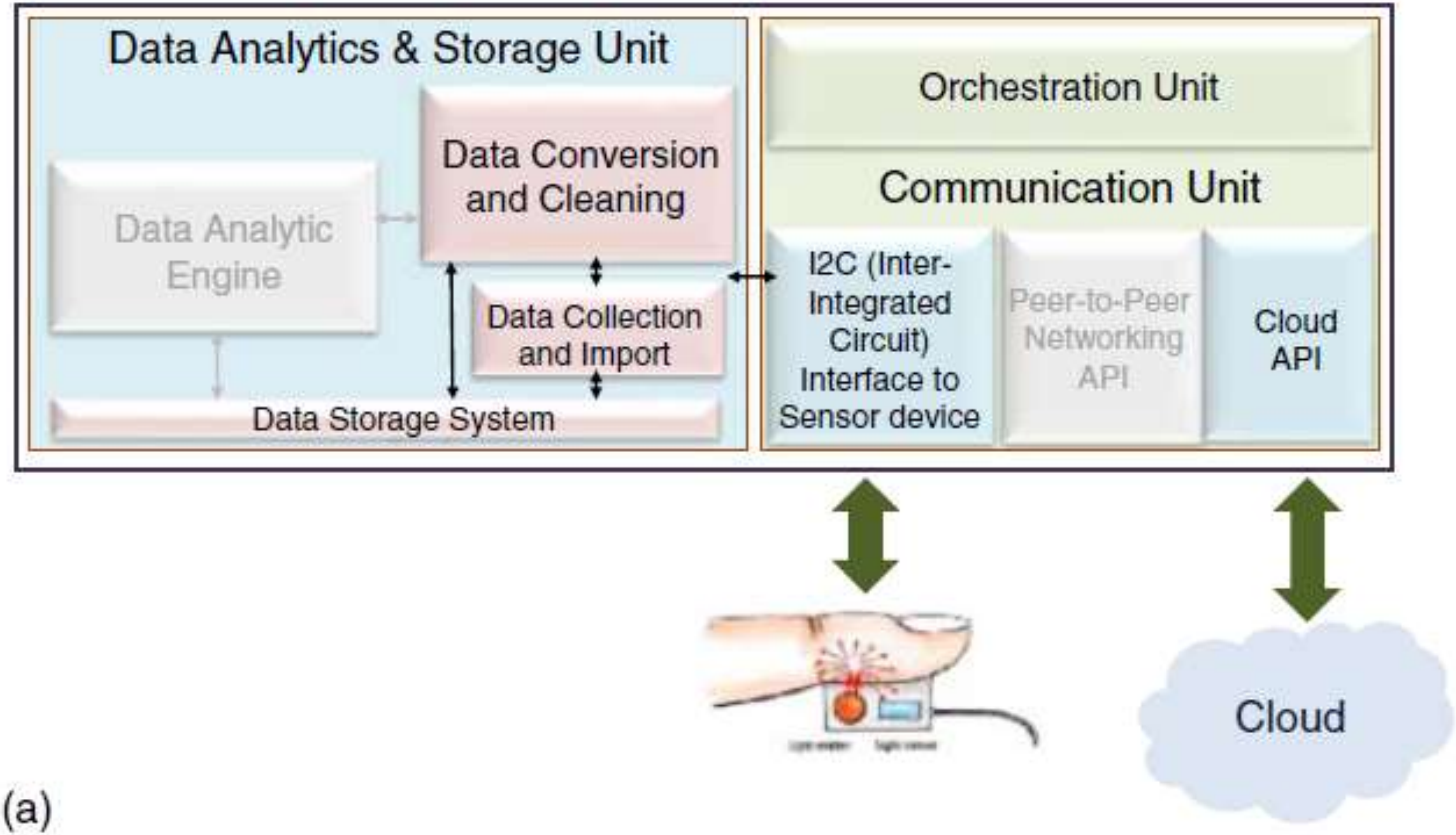
Figure 11.4 (a) General architecture of fog-engine; (b) A detailed architecture for the communication unit.

4.2 Configurations

- The fog-engine can be deployed in various configurations, which include different operational setups to enhance its functionality in processing and managing data.

4.2.1 Fog-Engine as a Broker

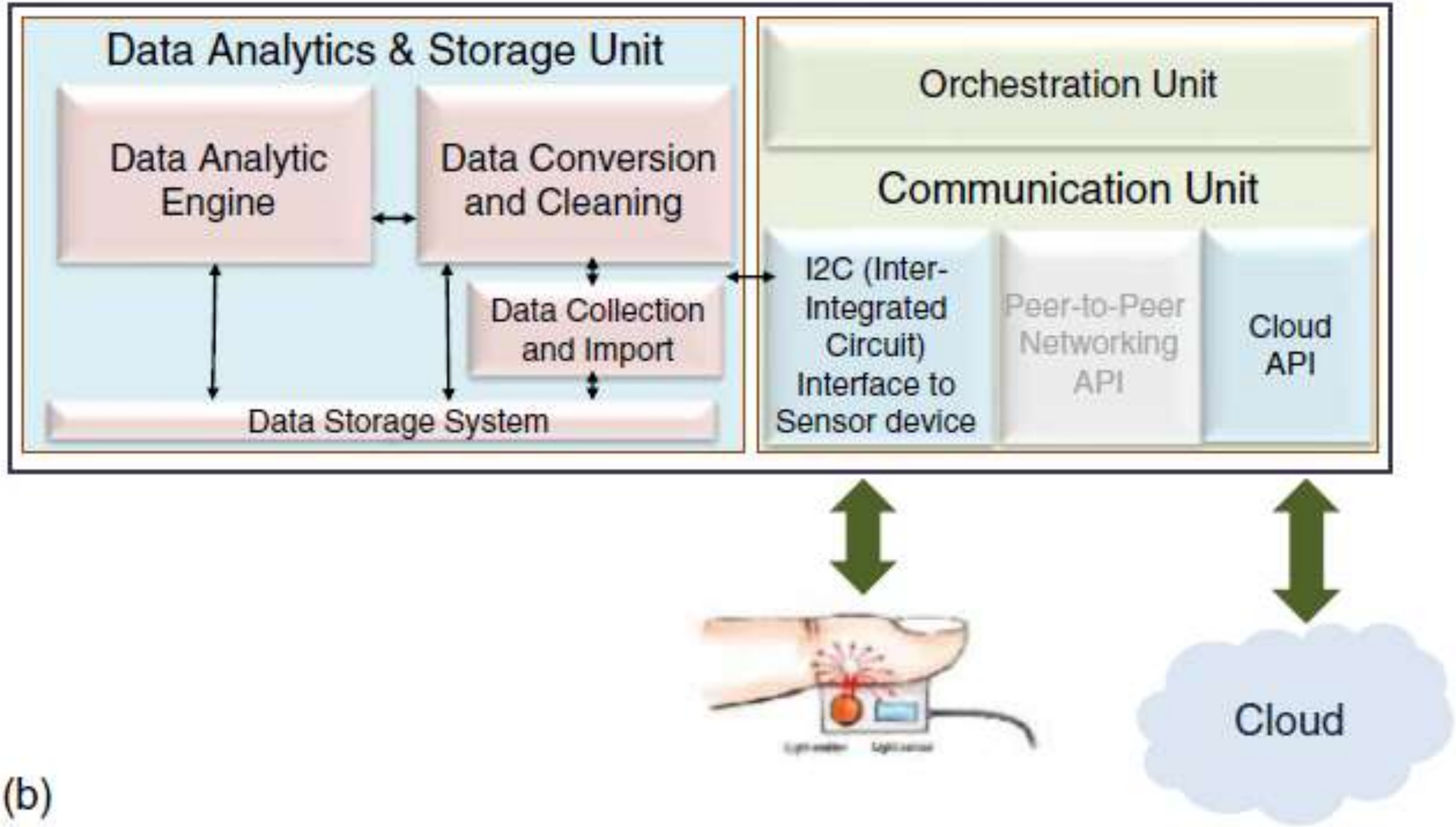
- **Role**: The fog-engine acts as an intermediary that manages data flow from sensors to the cloud.
- **Process**:
  - Receives data from sensors.
  - Filters and cleans the data before transmission.
  - Utilizes an I2C (Inter-Integrated Circuit) interface to connect and read from sensor devices.
- **Benefit**: This configuration reduces the amount of raw data sent to the cloud by preprocessing at the fog level.


Data Analytics & Storage Unit
Data Analytic Engine
Data Conversion and Cleaning
Data Collection and Import
Data Storage System
Orchestration Unit
Communication Unit
I2C (Inter-Integrated Circuit) Interface to Sensor device
Peer-to-Peer Networking API
Cloud API
Cloud

(a)

4.2.2 Fog-Engine as a Primary Data Analyzer

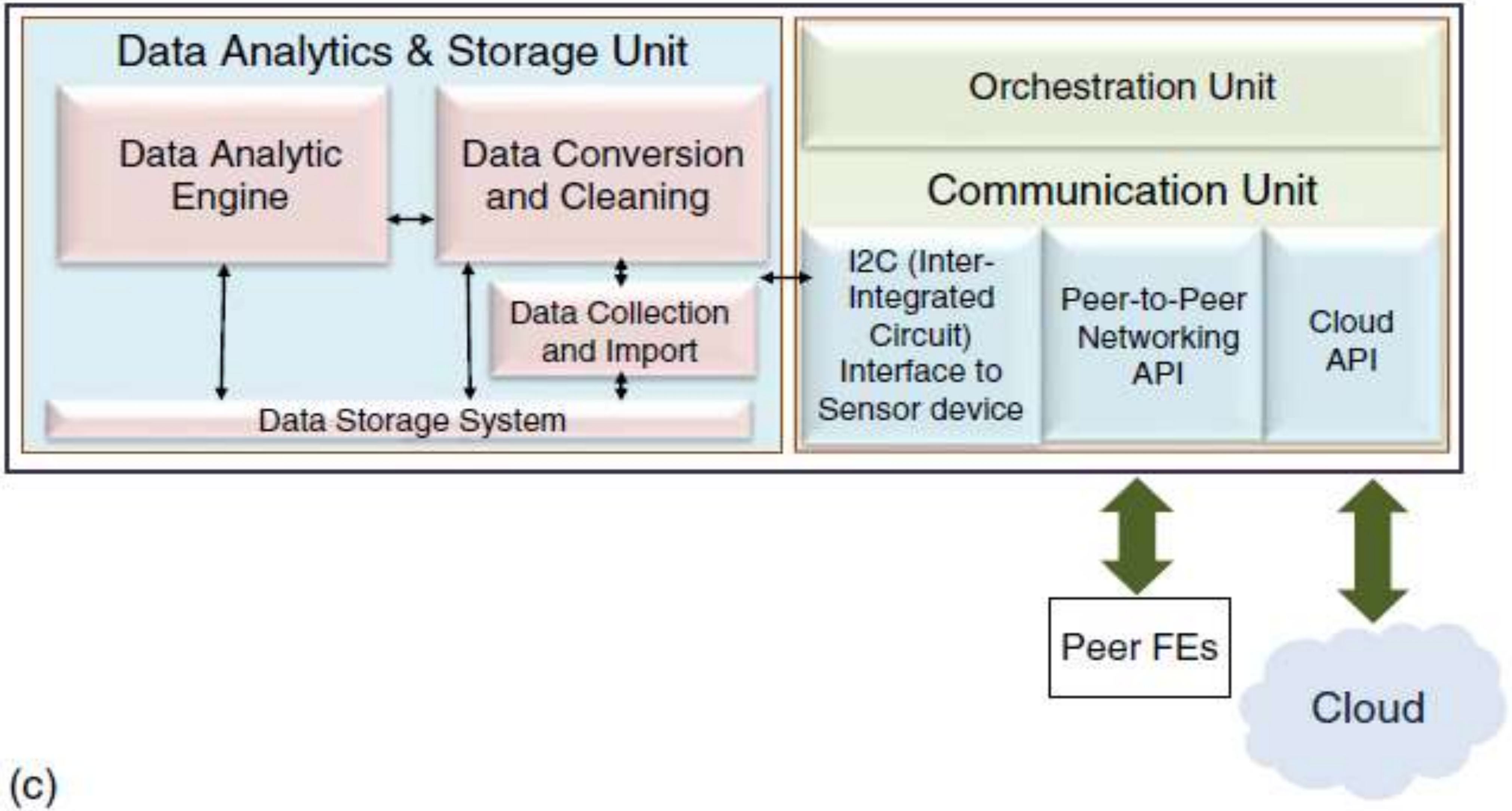
- **Functionality**: In this role, the fog-engine performs primary data analysis using algorithms and statistical methods before sending critical insights to the cloud.
- **Capabilities**:
    - Engages in local data processing which reduces bandwidth usage.
    - Provides immediate insights and responses based on real-time data analysis.


Data Analytics & Storage Unit
Data Analytic Engine
Data Conversion and Cleaning
Data Collection and Import
Data Storage System
Orchestration Unit
Communication Unit
I2C (Inter-Integrated Circuit) Interface to Sensor device
Peer-to-Peer Networking API
Cloud API
Cloud

(b)

4.2.3 Fog-Engine as a Server

- **Purpose**: Operates similar to a server that not only processes data but also serves multiple IoT devices.
- **Configuration**:
- Manages requests from various devices and performs operations like data aggregation and visualization.
- **Advantages**: This setup enhances resource allocation by facilitating data availability and analytics at the edge, thereby minimizing dependency on cloud resources.



(c)

Figure 11.5 Various configurations of fog-engine: (a) as a broker; (b) as a primary data analyzer; (c) as a server.

4.2.4 Communication with Fog-Engine versus the Cloud

Objective: This section explores the differences in communication strategies and efficiencies between using fog-engines and direct communication with the cloud.

Experimental Analysis:

- A series of experiments were conducted to evaluate the functionality and performance of fog-engines compared to cloud-based approaches.

Key Findings:

- **Data Transfer Efficiency**: Fog-engines allow for local data analysis, which significantly reduces the volume of data needing transmission to the cloud.
- **Reduced Data Volume**: The experiments revealed that utilizing fog-engines can decrease the size of transferred data by up to 40%.
- **Cost Efficiency**: With minimized data transmission to the cloud, there are associated reductions in costs related to cloud services.
- **Local Processing Advantages**:
  - Fog-engines perform local data analytics, which avoids the need for constant data streaming to the cloud, thus optimizing network usage and maintaining system efficiency.
  - This approach alleviates the complexities and costs associated with maintaining a steady connection with the cloud for data analytics.

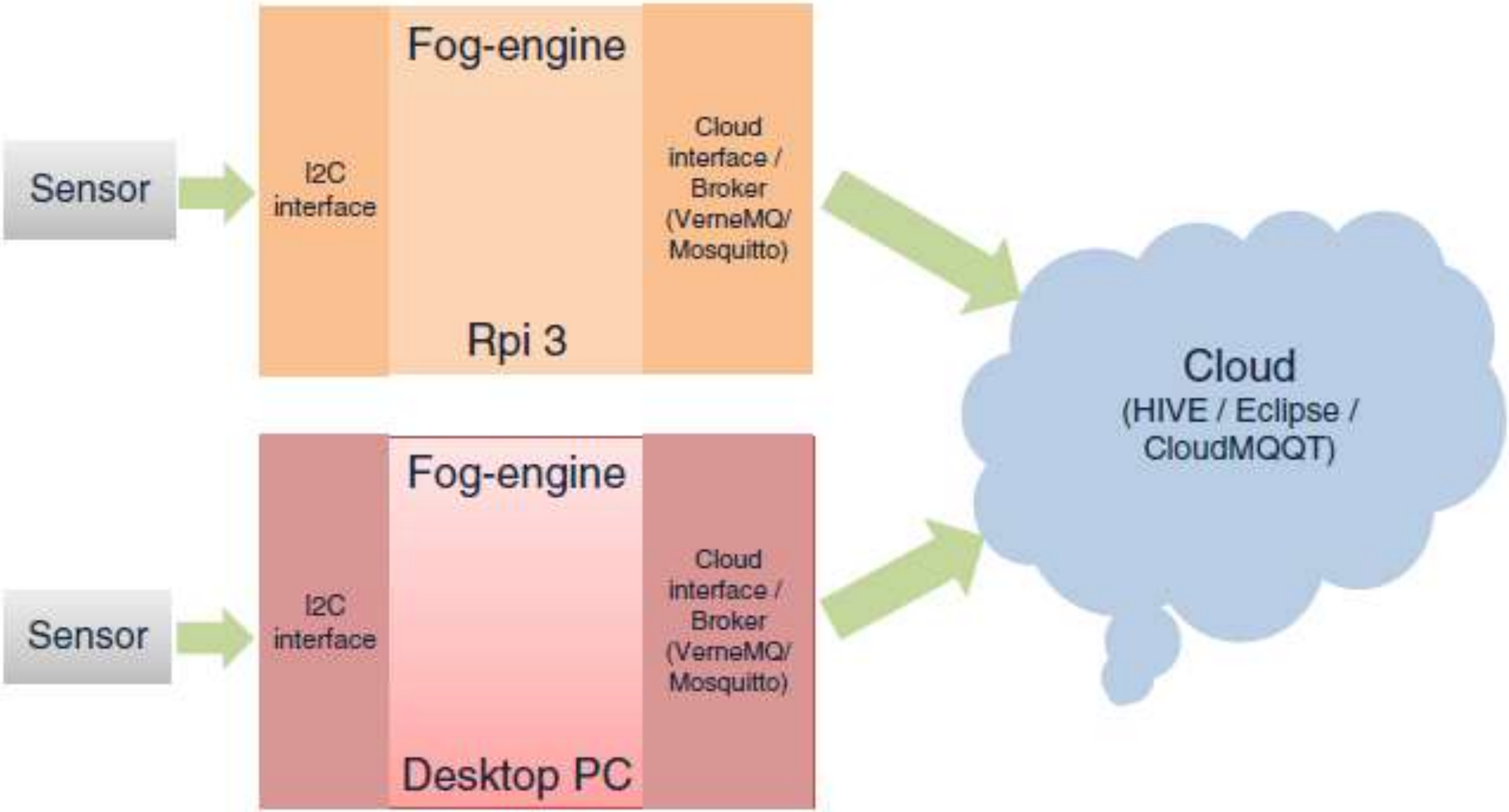


Figure 11.6 Fog-engine collects data and communicates with the cloud.

- The figure depicts a system where a fog-engine collects data from various sensors and communicates with the cloud.

Key Components:

- **Sensors**: Responsible for gathering data, such as environmental or operational metrics.
- **Fog-Engine**: Serves as an intermediary that preprocesses and analyzes incoming sensor data, ensuring only relevant and reduced data volumes are sent to the cloud.
- **Cloud**: Represents the central data analytics and storage system, receiving processed data from the fog-engine.